

❖ Prompt Engineering:

- ❖ **Prompt Meaning:** The word "prompt" can have several meanings depending upon the context in which it is used. Here are possible definitions :-
- ❖ (verb) To cause or encourage someone to do something; to spur them into action.
- ❖ Ex :- The teacher's reminder about the upcoming exam prompted the student to start studying.
- ❖ (adjective) Done without delay; on time, punctual
- ❖ Example: The prompt arrival of the ambulance saved the injured person life
- ❖ (noun) A cue or reminder that helps Someone remember what they are supposed to do or say.
- ❖ Ex :- The actor forget his lines, but the prompt from the stage manager helped him get back on the track.
- ❖ (noun) A message on screen that appears on a computer or other electronic device asking the user for input.
- ❖ Ex :- When I opened the program, a prompt appeared, asking me to enter my username and password.

❖ Below there are some prompt examples:

- **Purpose or Usage :** This prompt establishes a "meta-prompting" framework that forces the AI to think critically about how to solve a problem before actually writing the solution. We use it for several key reasons:
 1. **Overcomes Single-Mindedness:** AI often defaults to the most generic, conversational response. This prompt forces it to pause, evaluate the problem from multiple angles (e.g., a highly structured, analytical approach vs. a creative, role-play approach), and provide choices.
 2. **Improves Output Quality:** By separating the *strategy* (the alternative approaches) from the *execution* (the actual prompts), you ensure that the final prompts are deliberate, specialized, and highly optimized for different workflows.
 3. **Creates a Scalable Loop:** The inclusion of the concluding question ("Ask me for the next prompts...") automates the workflow, allowing you to seamlessly generate a library of diverse prompts back-to-back without having to redefine the rules every time.
- **Prompt:** Whenever I ask you to write a prompt for me to accomplish a task, List what the task is, list alternative approaches for completing the task and then write a prompt for yourself for each approach. When you're done ask me for the next prompts to create alternatives for.

Ask me what I want you to write a prompt for



Purpose or Usage: This meta-prompt transforms the AI into a collaborative editor. Instead of just answering a potentially vague or suboptimal query, it pauses to optimize the question first. This ensures that the eventual response is highly targeted, deeply contextual, and vastly superior to a standard, top-of-mind query.

- **Prompt:** When I ask a question, suggest me a better version for the question and ask me to use the question for better Results.



1. Persona Pattern

Purpose:

Assign a specific role to the AI so it responds from a particular perspective, expertise, or personality.

Why It Matters:

- Produces more relevant responses.
- Creates expert-level outputs.
- Improves consistency.

Example Prompt:

Act as a Senior Software Engineer with 10 years of experience.
Explain Object-Oriented Programming to a first-semester Computer Science student using simple examples.

Use Cases:

- Teacher
- Career Coach
- Software Engineer
- Marketing Expert
- Interviewer

2. Audience Persona Pattern

Purpose:

Define who the response is for.

Why It Matters:

The same topic should be explained differently to different audiences.

Example Prompt:

Explain Artificial Intelligence to:

1. A 10-year-old child
2. A university student
3. A business executive

Use language appropriate for each audience.

Use Cases:

- Education
 - Marketing
 - Technical Writing
 - Presentations
-

3. Flipped Interaction Pattern

Purpose:

Make the AI ask questions before answering.

Why It Matters:

- Collects missing information.
- Produces personalized responses.

Example Prompt:

Help me choose a programming language. Before making recommendations, ask me questions about my goals, interests, and experience level.

Use Cases:

- Career Guidance
 - Project Planning
 - Learning Paths
-

4. Question Refinement Pattern

Purpose:

Improve a vague question before answering.

Why It Matters:

Better questions lead to better answers.

Example Prompt:

Whenever I ask a question, first rewrite it into a clearer and more specific version. Then ask if I want to use the improved version before answering.

Use Cases:

- Research
 - Academic Work
 - Professional Communication
-

5. Cognitive Verifier Pattern

Purpose:

Force AI to verify reasoning before giving an answer.

Why It Matters:

Reduces mistakes and hallucinations.

Example Prompt:

Before answering, identify assumptions, verify logic, and check for possible errors. Then provide the final answer.

Use Cases:

- Mathematics
 - Programming
 - Decision Making
-

6. Outline Expansion Pattern

Purpose:

Generate a structure first, then expand each section.

Why It Matters:

Creates organized and comprehensive content.

Example Prompt:

Create an outline for learning Git and GitHub. After creating the outline, expand each section with explanations and examples.

Use Cases:

- Blogs
- Reports
- Study Notes
- Courses

7. Fact-Checking Techniques

Purpose:

Verify claims before accepting them.

Why It Matters:

Improves reliability and accuracy.

Example Prompt:

Review the following statement. Identify factual claims, verify them, highlight uncertainties, and provide corrections if necessary.

Statement:

"GitHub and Git are the same thing."

Use Cases:

- Research
 - Journalism
 - Academic Writing
-

8. Few-Shot Prompting

Purpose:

Teach AI using examples.

Why It Matters:

The model learns the desired pattern.

Example Prompt:

Example 1:

Input: Python

Output: Programming Language

Example 2:

Input: MySQL

Output: Database

Now classify:

Input: Git

Use Cases:

- Classification
- Formatting
- Content Generation

9. Prompt Design Best Practices

Purpose:

Create effective prompts consistently.

Key Principles:

- Be clear.
- Be specific.
- Define goals.
- Provide context.
- Specify format.
- Use examples.

Example Prompt:

Write a LinkedIn post announcing my Prompt Engineering certificate.

Requirements:

- Professional tone
 - 150 words
 - Include key concepts learned
 - End with a call to connect
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10. Structured Prompt Development**Purpose:**

Use a systematic prompt structure.

Common Framework:

Role

Task

Context

Requirements

Output Format

Example Prompt:

Role:

Career Coach

Task:

Review my LinkedIn profile summary.

Context:

I am a second-semester Computer Science student.

Requirements:

Provide strengths and improvements.

Output:

Bullet points.

11. Prompt Evaluation and Improvement Strategies**Purpose:**

Measure and improve prompt quality.

Evaluation Questions:

- Is the output accurate?
- Is it complete?
- Is it relevant?
- Is it clear?
- Is it actionable?

Example Prompt:

Evaluate the following prompt. Rate it from 1–10 for clarity, specificity, and effectiveness. Then suggest improvements.

12. AI-Assisted Problem Solving Techniques

Purpose:

Use AI as a thinking partner.

Why It Matters:

Breaks complex problems into manageable parts.

Example Prompt:

I want to build a GitHub portfolio.

Break this goal into:

1. Skills required
2. Projects required
3. Timeline
4. Resources
5. Weekly action plan

Use Cases:

- Project Planning
 - Learning Roadmaps
 - Business Decisions
-

13. Creative Prompting Methods

Purpose:

Generate innovative ideas.

Example Prompt:

Act as an inventor. Generate 20 unique AI project ideas for a Computer Science student that combine Prompt Engineering and GitHub.

Use Cases:

- Brainstorming
 - Entrepreneurship
 - Content Creation
-

14. Analytical Prompting Methods**Purpose:**

Encourage deep analysis.

Example Prompt:

Compare Git and GitHub.

Analyze:

- Purpose
- Advantages
- Limitations
- Real-world applications

Present findings in a table.

Use Cases:

- Research
 - Decision Making
 - Technical Analysis
-

15. Human-AI Collaboration Approaches**Purpose:**

Combine human creativity with AI capabilities.

Why It Matters:

The best results come from collaboration, not replacement.

Example Prompt:

I will provide my initial project idea.

Your role:

1. Improve the idea.
2. Identify weaknesses.
3. Suggest enhancements.
4. Create an implementation roadmap.

Wait for my idea first.

Use Cases:

- Project Development
- Writing
- Research
- Career Planning

16. Chain-of-Thought (CoT) Prompting**Purpose:**

Encourage the AI to reason through a problem step-by-step before providing the final answer.

Why It Matters:

- Improves reasoning quality.
- Helps solve complex problems.
- Reduces logical mistakes.

Example Prompt:

A store sells a notebook for \$5. Ali buys 3 notebooks and gives the cashier \$20. Think step-by-step and calculate how much change Ali should receive.

Use Cases:

- Mathematics
 - Programming
 - Logic Problems
 - Decision Making
-

17. Zero-Shot Prompting

Purpose:

Ask the AI to perform a task without providing examples.

Why It Matters:

- Fast and simple.
- Useful for common tasks.
- Requires less prompt writing.

Example Prompt:

Summarize the following article in 100 words.

Use Cases:

- Summarization
 - Translation
 - General Writing
-

18. Role-Based Prompting

Purpose:

Assign a specific professional role to the AI.

Why It Matters:

Different roles provide different perspectives.

Example Prompt:

Act as a cybersecurity expert. Explain common phishing attacks and how students can avoid them.

Use Cases:

- Education
 - Consulting
 - Technical Guidance
-

19. Meta Prompting

Purpose:

Use AI to create or improve prompts.

Why It Matters:

Helps users write better prompts and learn prompt engineering faster.

Example Prompt:

Create an advanced prompt that will help me learn GitHub as a beginner Computer Science student.

Use Cases:

- Prompt Optimization
 - Learning
 - Workflow Design
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20. Iterative Prompting

Purpose:

Improve outputs through multiple rounds of refinement.

Why It Matters:

The first answer is often not the best answer.

Example Prompt:

Round 1:

Write a LinkedIn post about completing a Prompt Engineering course.

Round 2:

Make it more professional.

Round 3:

Add a stronger conclusion and call-to-action.

Use Cases:

- Content Creation
- Resume Writing
- Project Planning

21. Constraint-Based Prompting

Purpose:

Force the AI to work within specific limitations.

Why It Matters:

Produces focused and controlled outputs.

Example Prompt:

Explain Git in less than 100 words. Use simple English and provide one real-world example.

Common Constraints:

- Word limits
 - Tone requirements
 - Output format
 - Reading level
-

22. Multi-Step Prompting

Purpose:

Break large tasks into smaller manageable stages.

Why It Matters:

Produces more organized and accurate outputs.

Example Prompt:

Help me create a GitHub portfolio.

Step 1: Identify portfolio sections.

Step 2: Suggest content.

Step 3: Create repository structure.

Step 4: Write README file.

Use Cases:

- Project Development
- Research

- Learning Plans
-

23. Reflection Prompting

Purpose:

Ask AI to review and critique its own response.

Why It Matters:

Improves quality and catches weaknesses.

Example Prompt:

Write a LinkedIn post about GitHub. After writing it, review your response and suggest three improvements.

Use Cases:

- Analysis
 - Writing
 - Problem Solving
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24. Self-Consistency Prompting

Purpose:

Generate multiple reasoning paths and select the most reliable answer.

Why It Matters:

Improves accuracy on complex reasoning tasks.

Example Prompt:

Solve this math problem using three different approaches. Compare the results and provide the most reliable answer.

Use Cases:

- Mathematics
- Research
- Strategic Decisions

25. Tree-of-Thought Prompting

Purpose:

Explore multiple possible solutions before choosing the best one.

Why It Matters:

Mimics human brainstorming and decision-making.

Example Prompt:

I want to earn money online as a Computer Science student. Generate three different career paths. Analyze advantages and disadvantages of each path. Recommend the best option.

Use Cases:

- Career Planning
 - Business Decisions
 - Project Selection
-

26. Re Act Prompting (Reason + Act)

Purpose:

Combine reasoning with actions.

Why It Matters:

Helps AI think, evaluate, and perform tasks systematically.

Example Prompt:

Analyze my GitHub repository. Reason about strengths and weaknesses. Then provide specific actions to improve it.

Use Cases:

- Research
 - Troubleshooting
 - Workflow Automation
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27. Socratic Prompting

Purpose:

Teach through questions rather than direct answers.

Why It Matters:

Develops critical thinking.

Example Prompt:

Help me understand Object-Oriented Programming. Do not explain directly. Instead, guide me by asking questions that help me discover the concepts myself.

Use Cases:

- Education
 - Coaching
 - Mentoring
-

28. Reverse Prompting

Purpose:

Work backward from a desired result.

Why It Matters:

Clarifies goals and requirements.

Example Prompt:

I want to become a freelance Prompt Engineer within one year. Work backward and create a month-by-month roadmap.

Use Cases:

- Goal Planning
 - Career Development
 - Project Management
-

29. Comparative Prompting

Purpose:

Compare multiple options systematically.

Why It Matters:

Supports informed decision-making.

Example Prompt:

Compare Python, Java, and C++ for Computer Science students.

Evaluate:

- Learning Difficulty
- Job Opportunities
- Community Support
- Project Development

Provide results in a table.

Use Cases:

- Technology Selection
 - Product Comparison
 - Career Decisions
-

30. Scenario-Based Prompting

Purpose:

Place AI inside a realistic situation.

Why It Matters:

Creates practical and contextual responses.

Example Prompt:

Imagine you are interviewing me for a junior software developer position. Conduct a realistic interview and provide feedback after each answer.

Use Cases:

- Interview Preparation

- Training
- Simulations